EXPERIMENT 2 : PART 1

RF COMPONENTS CHARACTERISTICS

SPA617M: RADIO LAB REPORT

by

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1. Aim

To experimentally characterize selected RF components used in an RF receiver chain, limited in this Part 1 report to the completed measurements of the RF filter, mixer, quarter-wave open-circuit stub, and filter input reflection.

2. Objectives Covered in Part 1

a) To characterize an RF filter, identify its type, and determine its cutoff frequency/passband.

b) To demonstrate a quarter-wave open-circuit stub as a notch filter.

c) To observe frequency conversion in an RF mixer and estimate its conversion loss.

d) To estimate the input reflection coefficient and return loss of the filter.

3. Apparatus Required

- RF signal generator(s)
- Spectrum analyser
- Digital storage oscilloscope
- RF low-pass filter / filter under test
- RF mixer with RF, LO and IF ports
- RF Tee connector
- Open-ended coaxial stub
- Short reference coaxial cable
- 50 Ω termination/load
- Suitable RF cables, adapters and attenuators

4. Theory

4.1 RF receiver-chain context

A minimal RF receiver chain uses filters to select/reject frequency ranges and a mixer to translate a signal from one frequency to another. The supplied laboratory manual illustrates the general receiver-chain arrangement.

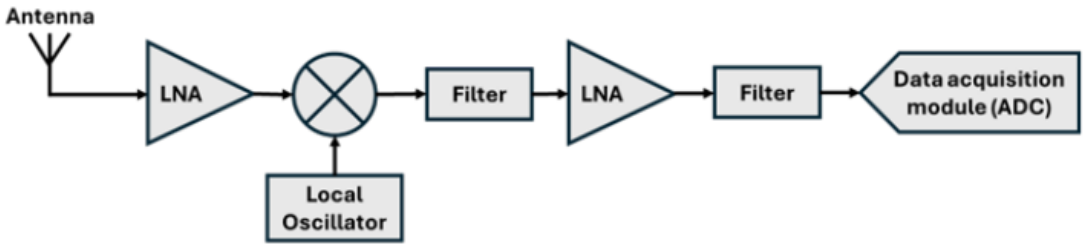


Figure 1. A minimal RF receiver chain.

Figure 1. Minimal RF receiver-chain arrangement from the supplied laboratory manual.

4.2 RF filter transmission characteristics

A low-pass filter passes frequencies below its cutoff frequency and attenuates higher frequencies.

The −3 dB cutoff frequency is measured relative to the minimum insertion loss in the passband. The observed response in this experiment decreases gradually and then falls sharply at the higher frequencies, identifying the device as a low-pass filter.

For low-pass and high-pass filters, fc denotes the cutoff frequency and fs denotes the stop band edge. For band-pass and notch filters, f1 and f2 denote the lower and upper cutoff frequencies, while f0 denotes the centre frequency or notch frequency. For the transmission measurement, let Pin be the power measured without the filter and Pout be the power measured after inserting the filter. The insertion loss is .

$$IL = Pin - Pout$$

A smaller value of IL indicates better transmission through the filter. The −3 dB cutoff frequencies are measured relative to the minimum insertion loss in the passband

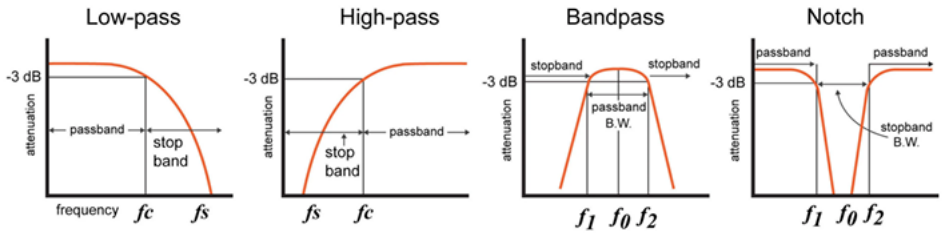


Figure 2. Typical transmission responses of common filter types

4.3 Mixer theory

An RF mixer is a three-port nonlinear device with RF, local-oscillator (LO), and intermediate-frequency (IF) ports. For RF frequency fRF and LO frequency f the principal mixing products observed here are the sum and difference:

$$f_{IF} = f_{RF} \pm f_{LO}$$

$$f_{IF} = f_{RF} - f_{LO} (for fRF > fLO)$$

The conversion loss is calculated from the RF input power PRF and the measured selected IF output power PIF:

$$Conversion\ loss = \ P_{RF} - P_{IF}(dB)$$

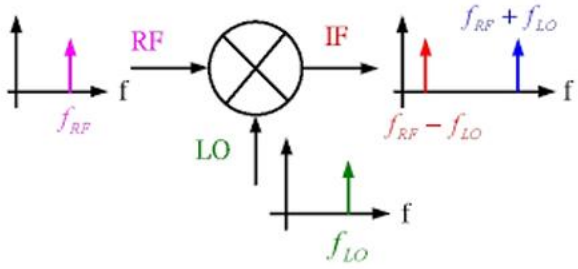


Figure 3. Operation of an RF mixer showing sum and difference frequency products.

4.4 Quarter-wave open-circuit stub

An open-ended coaxial stub connected in shunt through a Tee connector behaves as a resonator. At its quarter-wave resonant frequency, the open circuit at the far end is transformed approximately into a low impedance at the Tee junction, strongly attenuating the signal and producing a notch frequency =

$$f_0 = \frac{VF \cdot c}{(4L)}$$

Here L is the physical stub length, VF is the velocity factor, c is the speed of light in vacuum, and f0 is the expected notch frequency. The measured notch may differ slightly because of connector length and open-end effects.

4.5 Filter input reflection coefficient and return loss

When the filter input impedance is not equal to the characteristic impedance of the RF system, part of the incident signal is reflected. Using maximum and minimum voltage readings at the Tee junction:

$$|\Gamma| = \frac{(Vmax - Vmin)}{(Vmax + Vmin)}$$
$$RL = -20log10|\Gamma|(dB)$$

A smaller |Γ| indicates better matching, while a larger return loss indicates better matching. The Tee-connector method gives an approximate return-loss measurement because cable loss, connector mismatch and oscilloscope loading are not removed by calibration.

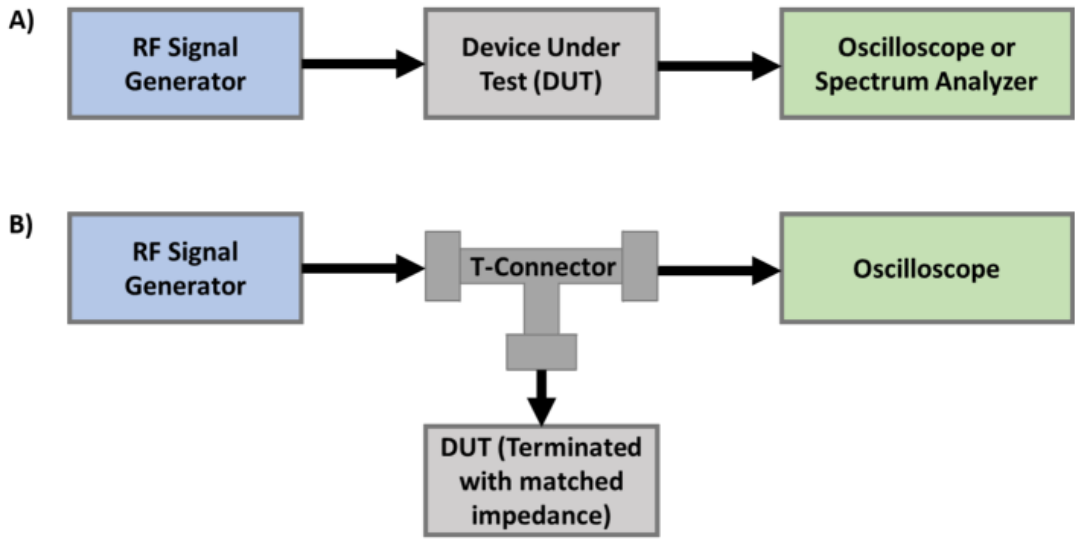


Figure 6. Experimental arrangements: (A) transmission measurement; (B) reflection measure-

Figure 4. Transmission and reflection measurement arrangements

5. Procedure

5.1 RF filter characterization

1. Connect the signal generator to the spectrum analyser through the short reference cable.
2. Set a suitable constant generator power.

3. Record the output/reference power at each test frequency without the filter and then insert the filter without changing the generator setting, cables or adapters.
4. Record the filter output power at the same frequencies.
5. Plot output power versus frequency and identify the filter type and approximate cutoff region.

5.2 Mixer characterization

6. Connect the RF and LO ports of the mixer to the signal sources and connect the IF port to the spectrum analyser.
7. Use the recorded LO frequency of 10 MHz and LO power of +7 dBm.
8. Apply the recorded RF frequencies of 13, 16 and 19 MHz with the recorded RF input power of +1.45 dBm.
9. Observe the sum (RF + LO) and difference (RF − LO) components on the spectrum analyser.
10. Record the measured IF powers and calculate conversion loss.

5.3 Quarter-wave stub experiment

11. Connect the signal generator to one port of the Tee connector and the spectrum analyser to the second port.
12. Connect the open-ended coaxial stub to the third port.
13. Measure the physical stub length and use the recorded velocity factor.
14. Calculate the expected notch frequency using $f0 = VF \cdot c/(4L)$ where VF is the velocity factor provided as 0.6954.
15. Observe the spectrum analyser and record the measured notch frequency.

5.4 Filter input reflection coefficient

16. Use the Tee-connector reflection measurement arrangement shown in Figure 4.
17. Set the oscilloscope input impedance to 1 MΩ and keep the generator amplitude constant.
18. Sweep a small frequency range around the recorded passband region.
19. Record the adjacent maximum and minimum voltage amplitudes.
20. Calculate $|\Gamma|$ from the maximum and minimum voltage values and calculate return loss.

6. Observation Tables

6.1 RF filter characterization

Frequency (MHz)	Recorded output power (dBm)
10	-1.06
50	-1.45
100	-1.57
150	-1.59
200	-1.71
250	-1.96
300	-2.16
350	-2.06
400	-2.56
450	-3.27
500	-3.33
550	-4.72
600	-7.68

650	-21.94
700	-31.39
750	-42.85
800	-36.86

6.2 Mixer observation

RF (MHz)	LO (MHz)	RF + LO (MHz)	Power (dBm)	RF − LO (MHz)	Power (dBm)
13	10	23	-7.00	3	-6.70
16	10	26	-7.18	6	-7.00
19	10	29	-6.87	9	-6.94

Recorded LO power: +7 dBm. Recorded RF input power: +1.45 dBm.

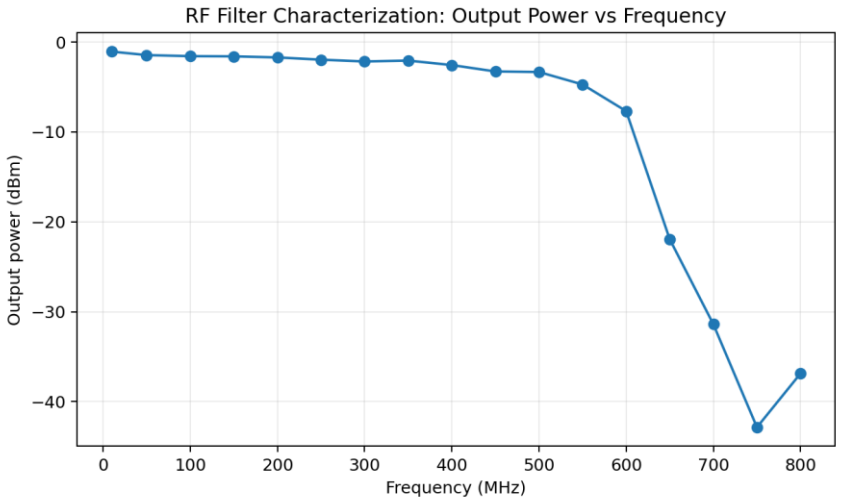
6.3 Quarter-wave stub observation

Parameter	Recorded value
Stub physical length, L	0.915 m (91.5 cm)
Velocity factor, VF	0.6954
Calculated notch frequency (notebook)	≈ 56.56 MHz
Measured notch frequency	57 MHz

6.4 Filter reflection / return-loss observation

Quantity	Recorded observation
Vmax	633 mV
Frequencies of Vmax	54 MHz and 68 MHz
Vmin	602 mV
Frequencies of Vmin	46 MHz and 60 MHz
|Γ|	0.0251
RL	32.01dB

7. Graphs



7.1 Filter output response

Figure 5. Recorded filter output power as a function of frequency.

The graph shows a relatively low attenuation at the lower frequencies followed by a strong reduction in output power above approximately the 500–550 MHz region.

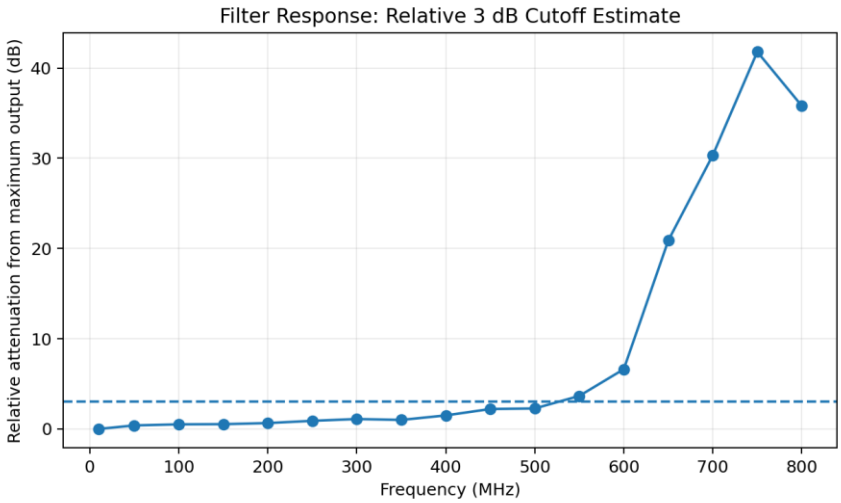


Figure 6. Relative attenuation referenced to the maximum recorded output

This is consistent with the observed low-pass response.

7.2 Relative 3 dB cutoff estimate

The −3 dB level relative to the maximum recorded output is reached between 500 MHz and 550 MHz. Linear interpolation between the two recorded points gives an approximate cutoff of 526.3 MHz as the estimated final value.

8. Calculations

8.1 Filter cutoff estimate

The maximum recorded output is −1.06 dBm at 10 MHz. A 3 dB reduction corresponds to:

$$P_{3dB} \approx -1.06 - 3 = -4.06dBm$$

The −4.06 dBm level lies between the 500 MHz point (−3.33 dBm) and the 550 MHz point (−4.72 dBm). Using linear interpolation:

$$fc \approx 500 + \frac{-4.06 + 3.33}{-4.72 + 3.33} \times 50 \approx 526.3MHz$$

Thus, the recorded response gives an approximate −3 dB cutoff near 526 MHz.

8.2 Mixer frequency conversion and conversion loss

For RF = 13 MHz and LO = 10 MHz: sum = 23 MHz, difference = 3 MHz. Using PRF = 1.45 dBm, conversion losses are (1.45 + 7) = 8.45 dB (sum product) and(1.45+6.70)= 8.15 dB (difference product).

For RF = 16 MHz and LO = 10 MHz: sum = 26 MHz, difference = 6 MHz. Using PRF = 1.45 dBm, conversion losses are(1.45+7.18)= 8.63 dB (sum product) and (1.45+7)=8.45 dB (difference product).

For RF = 19 MHz and LO = 10 MHz: sum = 29 MHz, difference = 9 MHz. Using PRF = 1.45 dBm, conversion losses are (1.45+6.87)=8.32 dB (sum product) and (1.45+6.94)=8.39 dB (difference product).

8.3 Quarter-wave stub notch frequency

$$f0 = \frac{VF \cdot c}{4\mathrm{L}}$$

$$f0 \approx \frac{(0.69)(3 \times 10^8)}{(4 \times 0.915)} \approx 56.56MHz$$

The notebook records the expected value as approximately 56.56 MHz and the measured notch as 57 MHz.

8.4 Filter reflection coefficient and return loss

Using the voltage values recorded in the notebook directly:

$$|\Gamma| = \frac{(633 - 602)}{(633 + 602)} = \frac{31}{1235} \approx 0.0251$$

$$RL = -20log10(0.0251) \approx 32.01dB$$

9. Result and Discussion

1. The RF filter was identified as a low-pass filter from the measured transmission response. The output remains comparatively high at low frequencies and decreases strongly at higher frequencies. The recorded data give an approximate −3 dB cutoff of 526 MHz by interpolation between 500 and 550 MHz.

2. Mixer frequency conversion was observed for RF frequencies of 13, 16 and 19 MHz with a fixed LO of 10 MHz. The expected sum products were 23, 26 and 29 MHz and the difference products were 3, 6 and 9 MHz, matching the frequencies recorded from the spectrum analyser. The corresponding conversion losses are approximately 8.15–8.63 dB for the recorded sum/difference components when PRF = +1.45 dBm is used.

3. For the quarter-wave open-circuit stub, a physical length of 0.915 m and a recorded velocity factor of 0.6954 were used. The calculation used VF ≈ 0.6954 and obtained 56.56 MHz, while the measured notch frequency was 57 MHz. The close agreement supports the quarter-wave-resonator behaviour.

4. The filter input reflection experiment recorded Vmax = 633 mV and Vmin = 602 mV at the listed frequencies. The notebook reports |Γ| = 0.0625 and RL = 24.08 dB. Direct substitution of the recorded voltages into the manual formula gives |Γ| ≈ 0.0251 and RL ≈ 32.0 dB, so the discrepancy should be checked against the original oscilloscope scale/readings before final submission.